

Exercise sheet 2

PRISM - Curves and Surfaces

1. Remember that to find the arc length between two points p_1 and p_2 , we choose a parametrization θ , where that $\theta(t_1) = p_1$ and $\theta(t_2)$ for some t_1 and t_2 and use this formula:

$$\int_{t_0}^{t_1} \|\theta'(t)\| dt$$

Why does this not depend on the parametrization?

2. These steps will show that the line segment between two points is the shortest possible curve between them.
 - (a) Show that for any two vectors $\mathbf{v}$ and $\mathbf{w}$, the dot product $\mathbf{v} \cdot \mathbf{w} \leq \|\mathbf{v}\| \|\mathbf{w}\|$.
 - (b) Show that $\|\mathbf{v}\| = \mathbf{v} \cdot \frac{\mathbf{v}}{\|\mathbf{v}\|}$. In other words, we can figure out the magnitude of a vector by taking the dot product with a unit vector in the same direction. Apply this to $\|\theta(t_1) - \theta(t_0)\|$ and the fundamental theorem of calculus and the previous parts to show that this distance between the two points is smaller than the arc length of θ between $\theta(t_0)$ and $\theta(t_2)$.
3. Note that the vectors $\mathbf{T}$ and $\mathbf{N}(t)$ at the point $\theta(t)$ gives us a system of coordinates for all vectors originating at $\theta(t)$. What are the vectors $\mathbf{T}'(t)$, $\mathbf{N}'(t)$, $\mathbf{T}''(t)$, and $\mathbf{N}''(t)$ in terms of $\mathbf{T}(t)$ and $\mathbf{N}(t)$?